

# The Human-AI Partnership Framework

Where human judgment belongs

## PCF 10.0 — Acquire, Construct, and Manage Assets

HITL Maturity Assessment — All Processes

|                                   |                             |                                 |
|-----------------------------------|-----------------------------|---------------------------------|
| L3 PROCESSES<br><b>22</b>         | L4 ACTIVITIES<br><b>64</b>  | M1 BASELINE H%<br><b>70%</b>    |
| M3 DESIGN INTENT H%<br><b>47%</b> | M5 LEADING H%<br><b>26%</b> | $\Delta$ M1 → M3<br><b>-23%</b> |

This document reports HITL (Human-in-the-Loop) maturity ratios for all of PCF 10.0 (Acquire, Construct, and Manage Assets) — 4 process groups, 22 Level 3 processes, and 64 Level 4 activities in total — at five maturity levels from M1 (Undesigned) through M5 (Leading), with M3 marking HITL Design Intent.

At M3 (HITL Design Intent), this domain's aggregate human-involvement ratio is **47%**, down from a **70%** undesigned (M1) baseline — a **-23-point** AI convergence opportunity. Mixed-pattern L3 rows reflect an equal-weighted rollup of their real L4 children rather than a force-classified single pattern.

### HITL Maturity Assessment

H% = Human involvement at each maturity level · M3 = HITL Design Intent · Full L1→L2→L3→L4 hierarchy

| PCF      | L1-Domain / L2-Process Group / L3-Process / L4-Activity           | Pattern     | M1  | M2  | M3 ★ | M4  | M5  | $\Delta$ M1→M3 |
|----------|-------------------------------------------------------------------|-------------|-----|-----|------|-----|-----|----------------|
| 10.0     | Acquire, Construct, and Manage Assets · L1 DOMAIN AGGREGATE       | —           | 70% | 60% | 47%  | 35% | 26% | -23%           |
| 10.1     | Plan and acquire assets · L2 AVG                                  | —           | 73% | 64% | 52%  | 41% | 32% | -21%           |
| 10.1.1   | Develop property strategy and long term vision                    |             | 90% | 85% | 75%  | 65% | 57% | -15%           |
| 10.1.1.1 | Confirm alignment of property requirements with business strategy | Decision    | 92% | 88% | 80%  | 72% | 65% | -12%           |
| 10.1.1.2 | Appraise the external environment                                 | Knowledge   | 85% | 78% | 65%  | 52% | 42% | -20%           |
| 10.1.1.3 | Determine build, buy, or lease decision                           | Decision    | 92% | 88% | 80%  | 72% | 65% | -12%           |
| 10.1.2   | Plan facility                                                     |             | 83% | 76% | 65%  | 54% | 45% | -18%           |
| 10.1.2.1 | Design facility                                                   | Knowledge   | 85% | 78% | 65%  | 52% | 42% | -20%           |
| 10.1.2.2 | Analyze budget                                                    | Knowledge   | 85% | 78% | 65%  | 52% | 42% | -20%           |
| 10.1.2.3 | Select property                                                   | Decision    | 92% | 88% | 80%  | 72% | 65% | -12%           |
| 10.1.2.4 | Negotiate terms for facility                                      | Decision    | 92% | 88% | 80%  | 72% | 65% | -12%           |
| 10.1.2.5 | Manage construction or modification to building                   | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.3   | Provide workspace and facilities                                  | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.3.1 | Acquire workspace and facilities                                  | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.3.2 | Change fit/form/function of workspace and facilities              | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.4   | Manage facilities operations                                      | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.4.1 | Relocate people                                                   | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.4.2 | Relocate material and tools                                       | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.1.4.3 | Provide facility access and security                              | Transaction | 60% | 48% | 35%  | 22% | 12% | -25%           |
| 10.2     | Design and construct assets · L2 AVG                              | —           | 68% | 58% | 46%  | 33% | 23% | -22%           |

## The Human-AI Partnership Framework

PCF 10.0 — HITL Maturity Assessment

| PCF           | L1-Domain / L2-Process Group / L3-Process / L4-Activity   | Pattern                                                                             | M1         | M2         | M3 ★       | M4         | M5         | Δ M1→M3     |
|---------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|-------------|
| <b>10.2.1</b> | <b>Manage capital program for assets</b>                  |    | <b>79%</b> | <b>71%</b> | <b>60%</b> | <b>49%</b> | <b>40%</b> | <b>-19%</b> |
| 10.2.1.1      | Define capital investment plan                            | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.2.1.2      | Monitor capital program                                   | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.1.3      | Secure construction financing                             | Decision                                                                            | 92%        | 88%        | 80%        | 72%        | 65%        | -12%        |
| <b>10.2.2</b> | <b>Design and plan asset construction</b>                 |    | <b>75%</b> | <b>66%</b> | <b>53%</b> | <b>40%</b> | <b>30%</b> | <b>-22%</b> |
| 10.2.2.1      | Develop construction strategy                             | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.2.2.2      | Perform construction performance management               | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.2.3      | Obtain construction permissions                           | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.2.4      | Design assets                                             | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.2.2.5      | Plan construction resources                               | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>10.2.3</b> | <b>Schedule and perform construction work</b>             | Transaction                                                                         | <b>60%</b> | <b>48%</b> | <b>35%</b> | <b>22%</b> | <b>12%</b> | <b>-25%</b> |
| 10.2.3.1      | Schedule construction work                                | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.3.2      | Obtain resources                                          | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.3.3      | Construct new assets                                      | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.3.4      | Augment existing assets                                   | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.3.5      | Renew/Replace assets                                      | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| <b>10.2.4</b> | <b>Manage asset construction</b>                          | Transaction                                                                         | <b>60%</b> | <b>48%</b> | <b>35%</b> | <b>22%</b> | <b>12%</b> | <b>-25%</b> |
| 10.2.4.1      | Monitor work performance                                  | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.4.2      | Undertake construction quality control                    | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.4.3      | Create work and asset records                             | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.4.4      | Manage safety, security, and access to sites              | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.2.4.5      | Schedule and perform commissioning                        | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| <b>10.3</b>   | <b>Maintain assets · L2 AVG</b>                           | —                                                                                   | <b>72%</b> | <b>62%</b> | <b>50%</b> | <b>37%</b> | <b>28%</b> | <b>-22%</b> |
| <b>10.3.1</b> | <b>Plan asset maintenance</b>                             |  | <b>80%</b> | <b>71%</b> | <b>57%</b> | <b>44%</b> | <b>34%</b> | <b>-23%</b> |
| 10.3.1.1      | Develop maintenance strategies                            | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.3.1.2      | Analyze assets and predict maintenance requirements       | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.3.1.3      | Specify maintenance policies                              | Document                                                                            | 72%        | 58%        | 40%        | 28%        | 18%        | -32%        |
| 10.3.1.4      | Integrate preventive maintenance into operations schedule | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.1.5      | Identify work management tasks & priorities               | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.3.1.6      | Conduct resource planning                                 | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.3.1.7      | Create asset work plans                                   | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>10.3.2</b> | <b>Manage asset maintenance</b>                           |  | <b>65%</b> | <b>54%</b> | <b>41%</b> | <b>29%</b> | <b>20%</b> | <b>-24%</b> |
| 10.3.2.1      | Schedule maintenance work                                 | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.2.2      | Obtain required resources                                 | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.2.3      | Undertake quality control                                 | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.2.4      | Update work and asset records                             | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.2.5      | Manage maintenance work safety                            | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.2.6      | Define maintenance performance targets                    | Decision                                                                            | 92%        | 88%        | 80%        | 72%        | 65%        | -12%        |
| 10.3.2.7      | Monitor maintenance performance against targets/contracts | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| <b>10.3.3</b> | <b>Perform asset maintenance</b>                          |  | <b>72%</b> | <b>63%</b> | <b>50%</b> | <b>39%</b> | <b>29%</b> | <b>-22%</b> |
| 10.3.3.1      | Perform preventative asset maintenance                    | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.3.2      | Perform routine asset maintenance                         | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.3.3      | Perform corrective asset maintenance and repairs          | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 10.3.3.4      | Identify unplanned maintenance requirements               | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 10.3.3.5      | Perform unplanned maintenance and repairs                 | Exception                                                                           | 95%        | 92%        | 82%        | 75%        | 68%        | -13%        |
| <b>10.4</b>   | <b>Manage asset end-of-life · L2 AVG</b>                  | —                                                                                   | <b>66%</b> | <b>55%</b> | <b>42%</b> | <b>29%</b> | <b>20%</b> | <b>-24%</b> |
| <b>10.4.1</b> | <b>Develop exit strategy</b>                              |  | <b>72%</b> | <b>63%</b> | <b>50%</b> | <b>37%</b> | <b>27%</b> | <b>-22%</b> |

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| PCF            | L1-Domain / L2-Process Group / L3-Process / L4-Activity       | Pattern                                                                           | M1  | M2  | M3 ★ | M4  | M5  | Δ M1→M3 |
|----------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|-----|------|-----|-----|---------|
| 10.4.1.1       | Monitor useful life of assets                                 | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.1.2       | Perform cost vs. benefit analysis for replacement             | Knowledge                                                                         | 85% | 78% | 65%  | 52% | 42% | -20%    |
| <b>10.4.2</b>  | <b>Decommission productive assets</b>                         | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.3</b>  | <b>Perform sale or trade</b>                                  | Decision                                                                          | 92% | 88% | 80%  | 72% | 65% | -12%    |
| <b>10.4.4</b>  | <b>Manage take-back centers</b>                               | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.5</b>  | <b>Dismantle assets</b>                                       | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.6</b>  | <b>Track parts</b>                                            | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.7</b>  | <b>Recycle parts</b>                                          | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.8</b>  | <b>Ship hazardous material</b>                                | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| <b>10.4.9</b>  | <b>Provide government reporting</b>                           | Document                                                                          | 72% | 58% | 40%  | 28% | 18% | -32%    |
| <b>10.4.10</b> | <b>Perform abandonment</b>                                    |  | 63% | 50% | 36%  | 24% | 14% | -27%    |
| 10.4.10.1      | Demolish and secure                                           | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.10.2      | Restore site                                                  | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.10.3      | Monitor site for duration of time required by regulators      | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.10.4      | Prepare handover to new operator or landowner                 | Document                                                                          | 72% | 58% | 40%  | 28% | 18% | -32%    |
| <b>10.4.11</b> | <b>Perform waste and hazardous goods management</b>           |  | 71% | 59% | 44%  | 32% | 22% | -27%    |
| 10.4.11.1      | Maintain material safety data sheets                          | Document                                                                          | 72% | 58% | 40%  | 28% | 18% | -32%    |
| 10.4.11.2      | Maintain awareness of and communicate regulatory requirements | Document                                                                          | 72% | 58% | 40%  | 28% | 18% | -32%    |
| 10.4.11.3      | Maintain hazardous material handling and disposal             | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.11.4      | Manage compliance audits                                      | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.11.5      | Recognize need to disposition hazardous materials/waste       | Knowledge                                                                         | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 10.4.11.6      | Determine disposition and associated processing               | Knowledge                                                                         | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 10.4.11.7      | Manage disposition, disposal, reprocessing activities         | Transaction                                                                       | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 10.4.11.8      | Manage documentation and reporting                            | Document                                                                          | 72% | 58% | 40%  | 28% | 18% | -32%    |

■ Decision 
 ■ Knowledge 
 ■ Document 
 ■ Transaction 
 ■ Exception

Full L1→L2→L3→L4 hierarchy · Mixed-pattern rows show an equal-weighted rollup of their real L4 children · © Timothy P. King & Claude (Anthropic) 2026